NOT MEASUREMENT SENSITIVE

MIL-F- 83828
26 October 1992

FILLERS AND SPACERS FOR ROTARY SWEEPER BRUSHES, FRONT MOUNTED OR TOWED TYPE SWEEPERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers steel wire fillers, polypropylene plastic fillers, steel spacers and plastic spacers which make up the brush assembly on rotary sweepers.

1.2 Classification.

1.2.1 Filler and spacer designations. Fillers (wafer and strip) and spacers covered by this specification shall be designated in the following form (see 6.2 and 6.6):

MILF 83828- XX X

Military Specification Code Number
Type of Filler or Spacer Code Number (see Table I and 1.2.2)
Class of Filler or Spacer Code Letter (see Table II and 1.2.3)

1.2.2 Type of filler or spacer. The type of filler (wafer or strip) or spacer is identified by a two digit number (see Table I).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Warner Robins Air Logistics Center, ATTN: WR-ALC/LVRG, 225 Ocmulgee Ct, Robins AFB GA, 31098-1647, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FSC-3825

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TABLE I. Code Number for Type of Filler or Spacer.

Code Number	Type of Filler or Spacer
01	Flat ring, continuous, steel filler (regular duty)
02	Flat ring, continuous, steel filler (heavy duty) (class A only)
03	Flat ring, continuous, plastic filler
04	Flat ring, continuous, steel spacer
05	Flat ring, continuous, plastic spacer
06	Flat ring, tufted, steel filler (regular duty)
09	Flat ring, tufted, steel spacer
10	Flat ring, tufted, plastic spacer
11	Convoluted ring, continuous, steel filler (regular duty) (class A only)
12	Convoluted ring, continuous, steel filler (heavy duty) (class A only)
13	Convoluted ring, continuous, plastic filler
14	Strip, tufted, steel filler (regular duty)
16	Strip, tufted, plastic filler

1.2.3 Class of filler or spacer. The class of filler (wafer or strip) or spacer shall be determined by its inside diameter (I.D.) and shall be identified by a single letter (see Table II).

TABLE II. Code Letter for Class of Filler or Spacer.

Code Letter	Class (inside diameter)
A	Class A (10 3/4 inch I.D.)
B	Class B (19 1/2 inch I.D.)
N	Not applicable (For tufted steel or plastic strips only)

2. APPLICABLE DOCUMENTS

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS
MILITARY
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-831 - Test Reports, Preparation of.

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation.

DEPARTMENT OF DEFENSE (DOD)
Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093).

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J403 - Chemical Compositions of SAE Carbon Steels.

(Application for copies should be addressed to Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale PA 15096).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Environmental conditions. The fillers shall satisfactorily withstand storage temperatures ranging from -60°F to +160°F and operating temperatures ranging from -40°F to +125°F.

3.2 Performance. When operated on concrete runways at a sweeping speed of 35 mph, a rotational speed of 500 revolutions per minute (rpm) and with a brush pattern of not less than 2 inches but not greater than 4 inches, the fillers shall be capable of operating for at least 200 hours. The loss of wire 2" or longer throughout the life of the filler shall not exceed 2 percent of the overall initial weight of the filler. The fillers shall maintain this performance at any operational temperature in the range specified in 3.1.

3.3 Construction.

3.3.1 Outside diameter. Fillers shall have an overall outside diameter (O.D.) tolerance of ± 1 inch. Flat and convoluted ring fillers shall have an overall O.D. (measured to the tip of the bristles) of 36 inches for class A and 46 inches for class B. The length of the bristles for strip fillers shall be as

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specified in paragraphs 3.3.4.1 and 3.3.4.2.

3.3.2 Continuous wafers. Each wafer shall consist of a steel ring filled with galvanized wire or polypropylene (plastic) bristles and shall be of the type and class as specified (see 6.2 b). Each wafer shall be marked with a black dot, or other suitable marking, on the ring to indicate the point of maximum static unbalance.

3.3.2.1 Ring. The ring shall be formed of AISI 1018 steel, in accordance with SAE J403, coil stock, 18 gauge minimum and shall be solid and continuous with no breaks to ensure structural integrity. The ring shall be either the flat or convoluted type as specified (see 6.2 b). Each ring shall have steel drive pins to engage the core. The pins shall have a minimum thickness of 5/16" for class A wafers and 1/4" for class B wafers. The length of the pins and number of pins per wafer shall be as specified under the individual wafer types below. Each ring shall be pressed closed every 1" ±1/8" to ensure maximum strength under high speed sweeping conditions. The ring shall have a thickness through the center of 15/32" ±1/16". The I.D. of a class A ring shall be 10 3/4" -0, +1/8", and the I.D. of a class B ring shall be 19 1/2" -0, +1/8". The O.D. of a class A ring shall be 12" -0, +1/8" for a flat type and 12 3/4" -0, +1/8" for a convoluted type. The O.D. of a class B ring shall be 20 3/4" -0, +1/8".

3.3.2.2 Bristles.

3.3.2.2.1 Steel wire bristles. Wire bristles shall be formed of AISI 1070 steel, in accordance with SAE J403, and shall be galvanized. The bristles shall be a minimum of 0.020" diameter for a class A wafer and a minimum of 0.018" diameter for a class B wafer. The bristles shall be crimped, with not less than 3 crimps per inch at a crimp amplitude of 1/16" minimum. The galvanized coating shall be applied at not less than 0.10 ounce per square foot, and the tensile strength of the wire shall be in the range of 340,000-350,000 pounds per square inch (psi) after crimping. The bristles shall be held secure in the ring by not less than 3 wraps of 0.090" minimum diameter AISI 1070 steel wire.

3.3.2.2.2 Plastic bristles. Plastic bristles shall be formed of virgin polypropylene strands having an oval cross section not less than 0.075" x 0.115". The strands shall be crimped, and the tensile strength shall be 35,000 psi minimum after crimping. The ring shall have a rubber liner to absorb friction and protect the bristles from abrasion at the ring edges. The bristles shall be held secure in the ring by not less than 3 wraps of 0.091" minimum diameter AISI 1070 steel wire.

3.3.2.3 Continuous wafer fillers (flat ring and convoluted).

3.3.2.3.1 Continuous, flat ring, steel wire wafer. The wafer shall be a flat ring filled with steel wire bristles as specified in 3.3.2.2.1. The class A wafer shall have at least two drive pins protruding a minimum of 9/16" from the ring and spaced at least 1 9/16" apart center to center. The class B wafer shall have not less than 4 drive pins. Each pin shall protrude at least 1/4" from the ring. The pins shall be 15 1/2" apart center to center. Weight of the complete wafer shall be 5.5 pounds (lbs) ±5 percent minimum for class A and 8.5 lbs ±5 percent minimum for class B.

3.3.2.3.2 Continuous, convoluted ring, steel wire wafer. The wafer shall be a convoluted ring filled with steel wire bristles as specified in 3.3.2.2.1. The

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wafer shall have at least two drive pins protruding at least 9/16" from the ring and spaced at least 1 9/16" apart center to center. Weight of the complete wafer shall be 5.5 lbs +5 percent minimum.

3.3.2.3.3 Continuous, flat ring, plastic wafer. The wafer shall be a flat ring filled with plastic bristles as specified in 3.3.2.2.2. The class A wafer shall have at least two drive pins protruding a minimum of 9/16" from the ring and spaced at least 1 9/16" apart center to center. The class B wafer shall have not less than 4 drive pins. Each pin shall protrude at least 1/4" from the ring. The pins shall be 15 1/2" apart center to center. Weight of the complete wafer shall be 4.5 lbs +5 percent minimum for class A and 8 lbs +5 percent minimum for class B.

3.3.2.3.4 Continuous, convoluted ring, plastic wafer. The wafer shall be a convoluted ring filled with plastic bristles as specified in 3.3.2.2.2. The wafer shall have at least two drive pins protruding at least 9/16" from the ring and spaced at least 1 9/16" apart center to center. Weight of the complete wafer shall be 4.6 lbs +5 percent minimum.

3.3.3 Tufted, steel wire wafer. The wafer shall be a flat ring filled with steel wire bristles. The ring shall be as specified in 3.3.2.1 except that it shall be formed of not less than 20 gauge strip steel. The wire shall be as specified in 3.3.2.2.1 and shall be formed into "tufts". Each tuft shall consist of not less than 120 single wires doubled over in a "U" shape to form a tuft with 240 sweeping ends. Individual tufts shall be inserted into a steel cup. The cups shall be formed as an integral part of the ring, one piece with driving lug attached, and there shall be not less than 16 cups per ring. The cup shall be formed in a manner to allow for brush "whip" action. A 0.115" minimum diameter pin will be inserted completely through the cup and "U" section of each tuft. The pin shall have a formed "head" at one end to prevent it from sliding through the cup and shall be bent at a 90° plane at the other end to secure the tuft in the cup. The cup shall be lined with rubber, and a damper ring formulated of the same material shall be molded to the liner to form a 1 piece unit. The damper ring shall be 1/2" width by 3/16" thickness minimum and shall have an inside diameter of 3/4" minimum. When installed on the tuft, the damper ring shall be 3 1/2" ±1/8" above the base.

3.3.4 Strip fillers.

3.3.4.1 Tufted, steel wire strip. The strip shall be formed of AISI 1070 steel channel 27 5/8" long by 1 5/8" wide by 2/16" thick minimum. The steel shall be in accordance with SAE J403. Each strip shall be filled with steel wire 0.018" minimum diameter, as specified in 3.3.2.2.1. The wire shall be formed into tufts as specified in 3.3.3 and shall have an 11 1/2" minimum bristle length. There shall be not less than 17 tufts per strip. Each tuft shall have an individual, one-piece, rubberized tubing cup and damper ring assembly as specified in 3.3.3. The channel shall be smooth, free from extrusions, retaining wires and protrusions for quick and easy installation. Each tuft shall be held on the strip by a separate retaining pin as specified in 3.3.3.

3.3.4.2 Tufted, plastic strip. The strip shall be formed as described in 3.3.4.1. Each strip shall be filled with plastic bristles as specified in 3.3.2.2.2. The bristles shall be crimped and formed into tufts with not less than 14 strands per tuft and with an 11 1/2" minimum bristle length. There shall be not less than 17 tufts per strip. Each tuft shall have an individual,

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one-piece, rubberized tubing cup and damper ring assembly as specified in 3.3.3. The tufts shall be held on the strip by a 12 gauge minimum AISI 1070 steel retaining wire, in accordance with SAE J403, through the base of each tuft along the length of the strip.

3.3.5 Heavy duty wafer fillers.

3.3.5.1 Continuous, flat ring, steel wire wafer, heavy duty. This wafer shall be identical to the class A wafer described in 3.3.2.3.1 except as follows:

a. The minimum diameter of the steel wire bristles shall be 0.018".

b. The bristles shall be held in the ring by not less than 4 wraps of retaining wire.

c. Weight of the complete wafer shall be 8 lbs ±5 percent minimum.

3.3.5.2 Continuous, convoluted ring, steel wire wafer, heavy duty. This wafer shall be identical to the class A wafer described in 3.3.2.3.2 except as follows:

a. The minimum diameter of the steel wire bristles shall be 0.018".

b. The bristles shall be held in the ring by not less than 4 wraps of retaining wire.

c. Weight of the complete wafer shall be 8 lbs ±5 percent minimum.

3.3.6 Spacers. Spacers for between wafer sections shall be plastic or AISI 1018 steel, in accordance with SAE J403 as specified (see 6.2 b). Spacers for continuous filled wafers shall be 1 3/16" thick ±1/16", and spacers for tufted wafers shall be 5/8" thick ±1/8". The I.D. of the spacers, and its tolerances, shall be the same as for the wafers to which they correspond (see 6.2 b). The O.D. of the spacers shall be 12 3/4" -0, +1/8" for class A wafers and 21 1/2 -0, +1/8" for class B.

3.4 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.2 d and 6.4).

3.5 Workmanship. The fillers shall be constructed and finished in accordance with the highest standards of the industry. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp corners and edges; accuracy of dimensions, pits, rust, etc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may use any inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete for a period of 2 years and be available to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that the product conforms to the prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the fillers shall be classified as follows:

a. First article testing (see 4.4 and 6.4).

b. Acceptance testing (see 4.5).

4.3 Test conditions. All testing shall be performed within an ambient temperature range of -40°F to +125°F, except the storage portion of the cold test. The requirements of each individual test specified herein shall be met.

4.4 First article testing.

4.4.1 First article test samples. The first article test sample shall consist of fillers identical to the items procured under the terms of the contract. Approval of the first article test report shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.4.2 First article tests. The first article tests shall consist of all tests listed under 4.6.

4.4.3 Test report. When required by the contract and upon completion of the first article tests, a first article test report shall be prepared in accordance with MIL-STD-831. The following shall be incorporated as a part of the test report:

a. All data collected, failures, maintenance performed, and other events recorded shall be identified by accumulated operation time, cycles, or position in the test procedure, as appropriate. Test conditions at the time of failures or irregular operations identified shall be recorded.

b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Summary of the engineering reasoning behind any corrections made, to be made, or proposed to be made and the predicted effectiveness of those corrections.

d. Test activity and/or contractor comments on item features or requirements that, if modified, should improve the item.

e. Test activity and/or contractor comments on field conditions or operating procedures to be avoided or cultivated to increase the reliability and useful life of the item.

4.5 Acceptance testing. Acceptance testing shall be classified as follows:

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a. Individual testing (see 4.5.1).

b. Random sample testing (see 4.5.2).

All defects identified shall be corrected prior to acceptance.

4.5.1 Individual test. Each filler shall be subjected to the examination of product test as described under 4.6.

4.5.2 Random sample testing. When specified in the contract or purchase order (see 6.2), any production filler may be selected by the procuring activity at any time to be subjected to the tests described in 4.6.1, 4.6.2 and 4.6.3. A certification in lieu of the test in 4.6.3 shall not be acceptable for random sample testing. Upon completion of the random sample testing, the contractor shall prepare a test report in accordance with MIL-STD-831.

4.5.2.1 Random sample test failure. When a filler fails to successfully complete any random sample test, the cause of the failure shall be determined, corrective action shall be taken, and the test shall be repeated to verify adequacy of the corrective action. Fillers to be delivered pursuant to the contract shall be inspected for all defects identified during all random sample tests and may not be accepted prior to verification of adequacy for the corrective action for a random sample test failure.

4.6 Test methods.

4.6.1 Examination of product. The fillers shall be inspected to determine compliance with the requirements specified in section 3 with respect to materials, workmanship and dimensions. For the first article and random sample items, this examination shall be accomplished by using a checklist which lists each requirement in section 3 that is not validated by tests in section 4. The results of this examination shall be noted beside each requirement and enclosed as part of the test report.

4.6.2 Wear tests. The tests shall be conducted on a smooth concrete or asphalt surface. The fillers shall be mounted on a broom core assembly, of the proper size and type, with the necessary spacers, if any are required, in accordance with the manufacturer's recommendations. The assembly need not be a full length core, but not less than 5 fillers shall be tested in the case of wafer type and not less than a sufficient number to provide tufts around the full circumference of the core in the case of strip type.

4.6.2.1 Series 1. The sweeping pattern shall be set at not less than 4 inches but not more than 6 inches. The core shall then be driven at not less than 500 rpm continuously for 8 hours minimum. The total wear of the bristles shall not exceed 4 percent. The percentage of total wear shall be determined as follows:

$$\frac{\text{(Filler Weight Before Test)} - \text{(Filler Weight After Test)}}{\text{Filler Weight Before Test}} \times 100$$

The percentage of total wear shall be determined for each filler. The shed bristle material, 6 inches in length or longer, shall not exceed 0.75 percent. The percentage of shed bristle material 6 inches or longer shall be determined as follows:

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$$\frac{\text{(Weight of Shed Material 6 Inches or Longer)}}{\text{Filler Weight Before Test}} \times 100$$

4.6.2.2 Series 2. Using the same fillers, the sweeping pattern shall then be set at not less than 2 inches but not more than 4 inches. The core shall be driven at not less than 500 rpm continuously for 8 hours minimum. Total wear and shed material shall be determined as specified in 4.6.2.1. The total wear of the bristles shall not exceed 1.5 percent. The shed material 6 inches or longer shall not exceed 0.20 percent.

4.6.3 Low temperature storage and operation. The fillers shall be tested in accordance with Test Method 502.2, Procedures I and II, of MIL-STD-810 to demonstrate compliance with 3.1. Operational tests shall be performed in accordance with paragraph 4.6.2. "Standard ambient" and "controlled ambient" shall be defined as the ambient condition of the test site. For Procedure I, the fillers shall be exposed to the minimum required cold storage temperature of 3.1 in a steady state condition for 24 hours. For Procedure II, the fillers shall be exposed to the minimum required cold operational temperature of 3.1 in a steady state condition. In lieu of actual cold chamber testing, the manufacturer may submit notarized certification that his product shall meet the requirements of this test.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the terms of the contract or purchase order.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers fillers that comprise the brush of snow sweepers used in the removal of new fallen snow, ice, slush or new snow from runways, taxiways and ramps, and other airfield paved areas. The sweeper can also be used for removing debris under high temperature conditions using plastic bristles.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this purchase description.

(b) Definitive military specification part number required (see 1.2 and 6.6).

(c) Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

(d) If first article testing is required (see 4.4 and 6.4).

(e) If random sample testing is required (see 4.5.2).

6.3 Consideration of data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements

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identified below shall be developed as specified by an approved Data Item Requirements List (CDRL), incorporated into the contract. When the provisions of DAR FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.
4.4.1, 4.4.2, 4.4.3 4.6.1, 4.6.2, 4.6.2.1, 4.6.2.2, 4.6.3.	First Article Test Report	DI-NDTI-80809A
4.5.2, 4.6.1, 4.6.2, 4.6.2.1, 4.6.2.2, 4.6.3.	Random Sample Test Report	DI-NDTI-80809A

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed in DOD 5000.19L, Vol II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Subject term (key word) listing.

Sweeper
Snow removal
Rotary brush
Wafer
Strip
Fillers

6.6 Definitive military specification part number. The military specification part number is a definitive part number which is formed by combining this specification number (MILF 83828) followed by a dash, the type of filler or spacer code number, a blank space and the class of filler or spacer code letter as shown in paragraph 1.2.1.

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MILITARY INTERESTS:

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Air Force - 99
Army - ME
Navy - YD

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Review Activities

Defense Logistics Agency - CS

PREPARING ACTIVITY:

Air Force - 84

Project No. 3825-0213

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